

1. Left join from employee e to employee m where the e.manager_id is equal to m.id

```
MariaDB [employee]> SELECT e.name AS employee, m.name AS manager FROM employees
LEFT JOIN employees m ON e.manager_id = m.id;
```

employee	manager
James	NULL
Maria	NULL
Robert	NULL
Emma	NULL
Liam	NULL
Olivia	NULL
Noah	NULL
Ava	NULL
William	NULL
Sophia	NULL
Ethan	NULL
Isabella	NULL
Charlotte	Emma
Ella	William
Scarlett	Olivia
Ava	Ethan
Scarlett	Sophia

2. Left join from departments to employees where manager id is equal to employee id

```
MariaDB [employee]> SELECT d.department_name, m.name
FROM employees m ON d.manager_id = m.id;
```

department_name	name
Assembly	James
Quality Control	Maria
Machining	Robert
Logistics	Emma
Maintenance	Liam
R&D	Olivia
Procurement	Noah
Health & Safety	Ava
Packaging	William
Inventory	Sophia
Welding	Ethan
Plant Management	Isabella

3. Left join from employees to departments where the department id is equal

```
MariaDB [employee]> SELECT e.name AS employee,
d.department_name AS department_name
FROM employees e LEFT JOIN departments d ON e.department_id =
d.department_id;
```

employee	department_name
James	Assembly
Maria	Quality Control
Robert	Machining
Emma	Logistics
Liam	Maintenance
Olivia	R&D
Noah	Procurement
Ava	Health & Safety
William	Packaging
Sophia	Inventory
Ethan	Welding
Isabella	Plant Management
Charlotte	Machining
Ella	Health & Safety
Scarlett	Health & Safety
Ava	Procurement
Scarlett	R&D

4. Where manager ID = null

```
MariaDB [emp]> SELECT * FROM employees
WHERE manager_id IS NULL;
```

name
James
Maria
Robert
Emma
Liam
Olivia
Noah
Ava
William
Sophia
Ethan
Isabella
Omar
Sherif

5. Where department ID = null

```
MariaDB [employee]> SELECT * FROM employees
WHERE department_id IS NULL;
```

name
Mia
Ava
Oliver
Omar
Aiden
Olivia
Logan
Mateo
Sherif
Sebastian
Ava
Charlotte

6. Used a left join from departments to employee getting the count of employee id's that have different department id's

```
MariaDB [employee]> SELECT d.department_name, COUNT(e.employee_id) AS employee_count
FROM departments d LEFT JOIN employees e ON d.department_id = e.department_id;
+-----+-----+
| department_name | employee_count |
+-----+-----+
| Assembly        | 16              |
| Quality Control  | 12              |
| Machining        | 13              |
| Logistics        | 13              |
| Maintenance      | 10              |
| R&D              | 15              |
| Procurement      | 13              |
| Health & Safety   | 24              |
| Packaging         | 11              |
| Inventory        | 16              |
| Welding          | 17              |
| Plant Management | 16              |
+-----+-----+
12 rows in set (0.002 sec)
```

7. Used a left join from employees to departments where department id's matched and where they dont have a manager (b/c they are the manager)

name	salary	department_name
James	95000	Assembly
Maria	92000	Quality Control
Robert	88000	Machining
Emma	91000	Logistics
Liam	85000	Maintenance
Olivia	110000	R&D
Noah	78000	Procurement
Ava	82000	Health & Safety
William	75000	Packaging
Sophia	84000	Inventory
Ethan	89000	Welding
Isabella	125000	Plant Management

8. Used a left join from employees e to employees m on manager e.manager_id = m.id and where e.salary > m.salary

employee	manager	employee_salary	manager_salary
Ella	James	97000	95000
Scarlett	William	77000	75000

2 rows in set (0.001 sec)